CHE 109: Engineering Chemistry – I
Summer-2025
Nuclear Chemistry
Chapter 4

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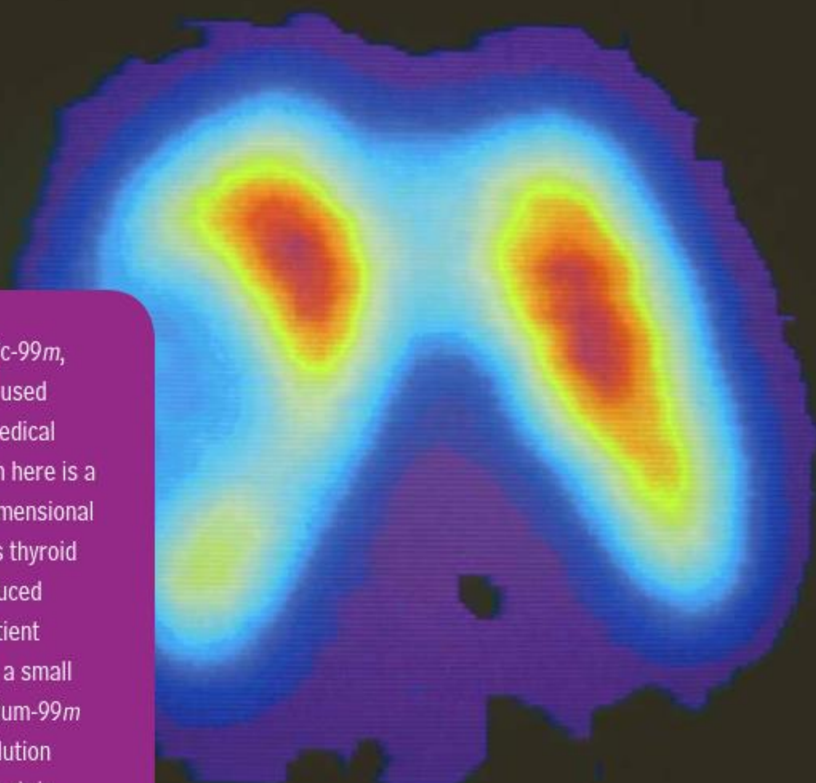
Nuclear Chemistry



Charing Cross Hospital/Science Source

Technetium-99*m*, Tc-99*m*, is the most widely used radioisotope for medical diagnostics. Shown here is a scintigram (two-dimensional scan) of a patient's thyroid gland. It was produced by injecting the patient intravenously with a small amount of technetium-99*m* in the form of a solution of sodium pertechnetate, $NaTcO_4$. Gamma rays emitted by the technetium were detected to produce the scintigram.

The Elements with Their Symbols and Atomic Masses*

Element	Symbol	Atomic Number	Atomic Mass†	Element	Symbol	Atomic Number	Atomic Mass†
Actinium	Ac	89	(227)	Mendelevium	Md	101	(256)
Aluminum	Al	13	26.98	Mercury	Hg	80	200.6
Americium	Am	95	(243)	Molybdenum	Mo	42	95.94
Antimony	Sb	51	121.8	Moscovium	Mc	115	(288)
Argon	Ar	18	39.95	Neodymium	Nd	60	144.2
Arsenic	As	33	74.92	Neon	Ne	10	20.18
Astatine	At	85	(210)	Neptunium	Np	93	(237)
Barium	Ba	56	137.3	Nickel	Ni	28	58.69
Berkelium	Bk	97	(247)	Nihonium	Nh	113	(284)
Beryllium	Be	4	9.012	Niobium	Nb	41	92.91
Bismuth	Bi	83	209.0	Nitrogen	N	7	14.01
Bohrium	Bh	107	(262)	Nobelium	No	102	(253)
Boron	B	5	10.81	Oganesson	Og	118	(294)
Bromine	Br	35	79.90	Osmium	Os	76	190.2
Cadmium	Cd	48	112.4	Oxygen	O	8	16.00
Calcium	Ca	20	40.08	Palladium	Pd	46	106.4
Californium	Cf	98	(249)	Phosphorus	P	15	30.97
Carbon	C	6	12.01	Platinum	Pt	78	195.1
Cerium	Ce	58	140.1	Plutonium	Pu	94	(242)
Cesium	Cs	55	132.9	Polonium	Po	84	(210)
Chlorine	Cl	17	35.45	Potassium	K	19	39.10
Chromium	Cr	24	52.00	Praseodymium	Pr	59	140.9
Cobalt	Co	27	58.93	Promethium	Pm	61	(147)
Copernicium	Cn	112	(285)	Protactinium	Pa	91	(231)
Copper	Cu	29	63.55	Radium	Ra	88	(226)
Curium	Cm	96	(247)	Radon	Rn	86	(222)
Darmstadtium	Ds	110	(269)	Rhenium	Re	75	186.2
Dubnium	Db	105	(260)	Rhodium	Rh	45	102.9
Dysprosium	Dy	66	162.5	Roentgenium	Rg	111	(272)
Einsteinium	Es	99	(254)	Rubidium	Rb	37	85.47
Erbium	Er	68	167.3	Ruthenium	Ru	44	101.1
Europium	Eu	63	152.0	Rutherfordium	Rf	104	(257)
Fermium	Fm	100	(253)	Samarium	Sm	62	150.4
Flerovium	Fl	114	(289)	Scandium	Sc	21	44.96
Fluorine	F	9	19.00	Seaborgium	Sg	106	(263)
Francium	Fr	87	(223)	Selenium	Se	34	78.96
Gadolinium	Gd	64	157.3	Silicon	Si	14	28.09
Gallium	Ga	31	69.72	Silver	Ag	47	107.9
Germanium	Ge	32	72.59	Sodium	Na	11	22.99
Gold	Au	79	197.0	Strontium	Sr	38	87.62
Hafnium	Hf	72	178.5	Sulfur	S	16	32.07
Hassium	Hs	108	(265)	Tantalum	Ta	73	180.9
Helium	He	2	4.003	Technetium	Tc	43	(99)
Holmium	Ho	67	164.9	Tellurium	Te	52	127.6
Hydrogen	H	1	1.008	Tennessine	Ts	117	(294)
Indium	In	49	114.8	Terbium	Tb	65	158.9
Iodine	I	53	126.9	Thallium	Tl	81	204.4
Iridium	Ir	77	192.2	Thorium	Th	90	232.0
Iron	Fe	26	55.85	Thulium	Tm	69	168.9
Krypton	Kr	36	83.80	Tin	Sn	50	118.7
Lanthanum	La	57	138.9	Titanium	Ti	22	47.88
Lawrencium	Lr	103	(257)	Tungsten	W	74	183.9
Lead	Pb	82	207.2	Uranium	U	92	238.0
Lithium	Li	3	6.941	Vanadium	V	23	50.94
Livermorium	Lv	116	(293)	Xenon	Xe	54	131.3
Lutetium	Lu	71	175.0	Ytterbium	Yb	70	173.0
Magnesium	Mg	12	24.31	Yttrium	Y	39	88.91
Manganese	Mn	25	54.94	Zinc	Zn	30	65.39
Meitnerium	Mt	109	(266)	Zirconium	Zr	40	91.22

*All atomic masses have four significant figures. These values are recommended by the Committee on Teaching of Chemistry, International Union of Pure and Applied Chemistry.
†Approximate values of atomic masses for radioactive elements are given in parentheses.

Nuclear chemistry is the study of reactions involving changes in atomic nuclei. This branch of chemistry began with the discovery of natural radioactivity by Antoine Becquerel and grew as a result of subsequent investigations by Pierre and Marie Curie and many others. Nuclear chemistry is very much in the news today. In addition to applications in the manufacture of atomic bombs, hydrogen bombs, and neutron bombs, even the peaceful use of nuclear energy has become controversial, in part because of safety concerns about nuclear power plants and also because of problems with radioactive waste disposal. In this chapter, we will study nuclear reactions, the stability of the atomic nucleus, radioactivity, and the effects of radiation on biological systems.

► Radioactivity and Nuclear Bombardment Reactions

In chemical reactions, only the outer electrons of the atoms are disturbed. The nuclei of the atoms are not affected. In nuclear reactions, however, the nuclear changes that occur are independent of the chemical environment of the atom. For example, the nuclear changes in a radioactive $^{3}_{1}H$ atom are the same if the atom is part of an H_2 molecule or incorporated into H_2O.

We will look at two types of nuclear reactions. One type is **radioactive decay**, *the process in which a nucleus spontaneously disintegrates, giving off radiation.* The radiation consists of one or more of the following, depending on the nucleus: electrons, nuclear particles (such as neutrons), smaller nuclei (usually helium-4 nuclei), and electromagnetic radiation.

The second type of nuclear reaction is a **nuclear bombardment reaction**, *a nuclear reaction in which a nucleus is bombarded, or struck, by another nucleus or by a nuclear particle.* If there is sufficient energy in this collision, the nuclear particles of the reactants rearrange to give a product nucleus or nuclei. First, we will look at radioactive decay.

Radioactivity

In 1895 the German physicist Wilhelm Röntgen[§] noticed that cathode rays caused glass and metals to emit very unusual rays. This highly energetic radiation penetrated matter, darkened covered photographic plates, and caused a variety of substances to fluoresce. Because these rays could not be deflected by a magnet, they could not contain charged particles as cathode rays do. Röntgen called them X rays because their nature was not known.

Not long after Röntgen's discovery, Antoine Becquerel,[†] a professor of physics in Paris, began to study the fluorescent properties of substances. Purely by accident, he found that exposing thickly wrapped photographic plates to a certain uranium compound caused them to darken, even without the stimulation of cathode rays. Like X rays, the rays from the uranium compound were highly energetic and could not be deflected by a magnet, but they differed from X rays because they arose spontaneously. One of Becquerel's students, Marie Curie,[‡] suggested the name ***radioactivity*** to describe this *spontaneous emission of particles and/or radiation.* Since then, any element that spontaneously emits radiation is said to be *radioactive.*

Three types of rays are produced by the *decay,* or breakdown, of radioactive substances such as uranium. Two of the three are deflected by oppositely charged metal plates (Figure 2.6). ***Alpha (α) rays*** consist of *positively charged particles,* called ***α particles,*** and therefore are deflected by the positively charged plate. ***Beta (β) rays,*** or ***β particles,*** are electrons and are deflected by the negatively charged plate. The third type of radioactive radiation consists of high-energy rays called ***gamma (γ) rays.*** Like X rays, γ rays have no charge and are not affected by an external field.

[§]Wilhelm Konrad Röntgen (1845–1923). German physicist who received the Nobel Prize in Physics in 1901 for the discovery of X rays.

[†]Antoine Henri Becquerel (1852–1908). French physicist who was awarded the Nobel Prize in Physics in 1903 for discovering radioactivity in uranium.

‡Marie (Marya Sklodowska) Curie (1867–1934). Polish-born chemist and physicist. In 1903 she and her French husband, Pierre Curie, were awarded the Nobel Prize in Physics for their work on radioactivity. In 1911, she again received the Nobel prize, this time in chemistry, for her work on the radioactive elements radium and polonium. She is one of only three people to have received two Nobel prizes in science. Despite her great contribution to science, her nomination to the French Academy of Sciences in 1911 was rejected by one vote because she was a woman! Her daughter Irene, and son-in-law Frederic Joliot-Curie, shared the Nobel Prize in Chemistry in 1935.

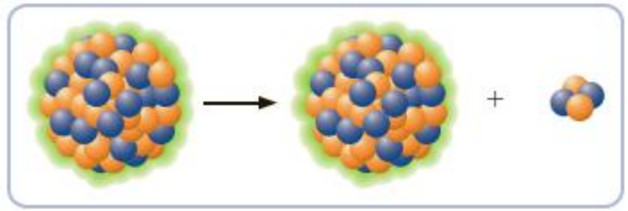
Nuclear Equations

Nuclide symbols were introduced in Section 2.3. For uranium-238, you write

mass number → 238
atomic number → 92

$^{238}_{92}\mathrm{U}$

You can write an equation for the nuclear reaction corresponding to the decay of uranium-238 much as you would write an equation for a chemical reaction. You represent the uranium-238 nucleus by the *nuclide symbol* $^{238}_{92}\mathrm{U}$. ◀ The radioactive decay of $^{238}_{92}\mathrm{U}$ by alpha-particle emission (loss of a $^{4}_{2}\mathrm{He}$ nucleus) is written



$$^{238}_{92}\mathrm{U} \longrightarrow {}^{234}_{90}\mathrm{Th} + {}^{4}_{2}\mathrm{He}$$

The product, in addition to being a helium-4 alpha particle, is thorium-234. This is an example of a **nuclear equation**, which is *a symbolic representation of a nuclear reaction.* Normally, only the nuclei are represented. It is not necessary to indicate the chemical compound or the electron charges for any ions involved, because the chemical environment has no effect on nuclear processes.

Reactant and product nuclei are represented in nuclear equations by their nuclide symbols. Other particles are given the following symbols, in which the subscript equals the charge and the superscript equals the total number of protons and neutrons in the particle (mass number):

Proton	$^{1}_{1}\mathrm{H}$	or	$^{1}_{1}\mathrm{P}$
Neutron	$^{1}_{0}\mathrm{n}$		
Electron	$^{0}_{-1}\mathrm{e}$	or	$^{0}_{-1}\beta$
Positron	$^{0}_{1}\mathrm{e}$	or	$^{0}_{1}\beta$
Gamma photon	$^{0}_{0}\gamma$		

The decay of a nucleus with the emission of an electron, $^{0}_{-1}\mathrm{e}$, is usually called beta emission, and the emitted electron is sometimes labeled $^{0}_{-1}\beta$. A **positron** is *a particle similar to an electron, having the same mass but a positive charge.* A **gamma photon** is *a particle of electromagnetic radiation of short wavelength (about 1 pm, or 10^{-12} m) and high energy.*

Example 20.2 Deducing a Product or Reactant in a Nuclear Equation

Gaining Mastery Toolbox

Critical Concept 20.2

The total charge (sum of all charges) and total number of nucleons (protons and neutrons) is conserved during a nuclear reaction. In a balanced nuclear reaction, the sum of all reactant superscripts must equal the sum of all product superscripts, and the sum of all reactant subscripts must equal the sum of all product subscripts.

Technetium-99 is a long-lived radioactive isotope of technetium. Each nucleus decays by emitting one beta particle. What is the product nucleus?

Problem Strategy First, write the nuclear equation using the symbol ${}^{A}_{Z}X$ for the unknown nuclide or particle. Then, solve for A and Z using the following relations: (1) the sum of the subscripts for the reactants equals the sum of the subscripts for the products, and (2) the sum of the superscripts for the reactants equals the sum of the superscripts for the products.

Solution Technetium-99 has the nuclide symbol ${}^{99}_{43}Tc$. A beta particle is an electron; the symbol is ${}^{0}_{-1}e$. The nuclear equation is

$${}^{99}_{43}Tc \longrightarrow {}^{A}_{Z}X + {}^{0}_{-1}e$$

(continued)

Example 20.2 *(continued)*

Solution Essentials:

- Nucleon
- Nuclear particles: proton (${}^{1}_{1}H$), neutron (${}^{1}_{0}n$), beta (${}^{0}_{-1}e$), positron (${}^{0}_{1}e$), gamma photon (${}^{0}_{0}\gamma$), and alpha (${}^{4}_{2}He$)
- Nuclear equation
- Nuclide symbol

From the superscripts, you can write

$$99 = A + 0, \text{ or } A = 99$$

Similarly, from the subscripts, you get

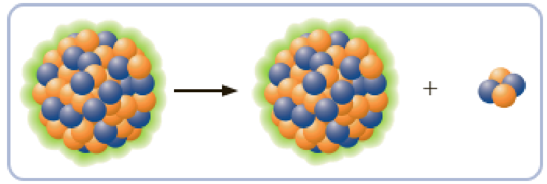
$$43 = Z - 1, \text{ or } Z = 43 + 1 = 44$$

Hence $A = 99$ and $Z = 44$, so the product is ${}^{99}_{44}X$. Because element 44 is ruthenium, symbol Ru, you write the product nucleus as $\mathbf{{}^{99}_{44}Ru}$.

Types of Radioactive Decay

There are six common types of radioactive decay; the first five are listed in Table 20.2. Note how the last column in Table 20.2 indicates the conditions where one would expect to observe a particular type of emission.

1. **Alpha emission** (abbreviated α): *emission of a $^{4}_{2}He$ nucleus, or alpha particle, from an unstable nucleus.* An example is the radioactive decay of radium-226.



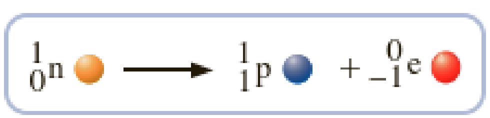
$$^{226}_{88}Ra \longrightarrow {}^{222}_{86}Rn + {}^{4}_{2}He$$

Table 20.2 Types of Radioactive Decay

			Resulting Nuclear Change		
Type of Decay	Radiation	Equivalent Process	Atomic Number	Mass Number	Usual Nuclear Condition
Alpha emission (α)	$^{4}_{2}He$	—	-2	-4	$Z > 83$
Beta emission (β)	$^{0}_{-1}e$	$^{1}_{0}n \longrightarrow {}^{1}_{1}p + {}^{0}_{-1}e$	$+1$	0	N/Z too large
Positron emission (β^{+})	$^{0}_{1}e$	$^{1}_{1}p \longrightarrow {}^{1}_{0}n + {}^{0}_{1}e$	-1	0	N/Z too small
Electron capture (EC)	X rays	$^{1}_{1}p + {}^{0}_{-1}e \longrightarrow {}^{1}_{0}n$	-1	0	N/Z too small
Gamma emission (γ)	$^{0}_{0}\gamma$	—	0	0	Excited

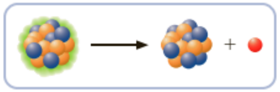
The product nucleus has an atomic number that is two less, and a mass number that is four less, than that of the original nucleus.

2. **Beta emission** (abbreviated β or β^{-}): *emission of a high-speed electron from an unstable nucleus.* Beta emission is the result of the conversion of a neutron to a proton.



$$^{1}_{0}n \longrightarrow {}^{1}_{1}p + {}^{0}_{-1}e$$

An example of beta emission is the radioactive decay of carbon-14.



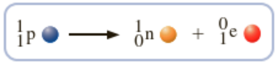
$$^{14}_{6}\mathrm{C} \longrightarrow {}^{14}_{7}\mathrm{N} + {}^{0}_{-1}\mathrm{e}$$

The product nucleus has an atomic number that is one more than that of the original nucleus. The mass number remains the same.

3. **Positron emission** (abbreviated β^+): *emission of a positron from an unstable nucleus.* A positron, denoted in nuclear equations as ${}^{0}_{1}\mathrm{e}$, is a particle identical to an electron in mass but having a positive instead of a negative charge. Positron emission is the result of the conversion of a proton to a neutron. ◀

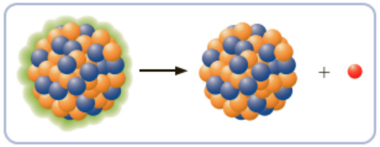
Positrons are annihilated as soon as they encounter electrons. When a positron and an electron collide, both particles vanish with the emission of two gamma photons that carry away the energy.

$${}^{0}_{1}\mathrm{e} + {}^{0}_{-1}\mathrm{e} \longrightarrow 2\,{}^{0}_{0}\gamma$$



$${}^{1}_{1}\mathrm{p} \longrightarrow {}^{1}_{0}\mathrm{n} + {}^{0}_{1}\mathrm{e}$$

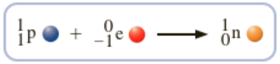
The radioactive decay of technetium-95 is an example of positron emission.



$${}^{95}_{43}\mathrm{Tc} \longrightarrow {}^{95}_{42}\mathrm{Mo} + {}^{0}_{1}\mathrm{e}$$

The product nucleus has an atomic number that is one less than that of the original nucleus. The mass number remains the same.

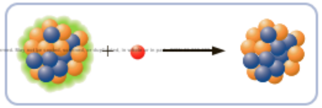
4. **Electron capture** (abbreviated EC): *the decay of an unstable nucleus by capturing, or picking up, an electron from an inner orbital of an atom.* In effect, a proton is changed to a neutron, as in positron emission.



$${}^{1}_{1}\mathrm{p} + {}^{0}_{-1}\mathrm{e} \longrightarrow {}^{1}_{0}\mathrm{n}$$

An example is given by potassium-40, which has a natural abundance of 0.012%. ◀ Potassium-40 can decay by electron capture, as well as by beta and positron emissions. The equation for electron capture is

Most of the argon in the atmosphere is believed to have resulted from the radioactive decay of ${}^{40}_{19}\mathrm{K}$.



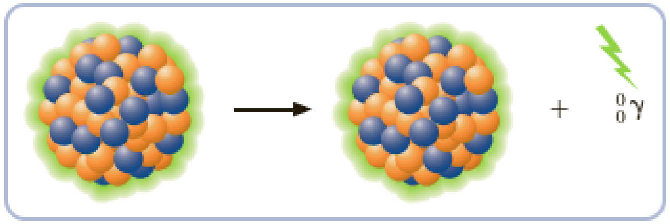
$${}^{40}_{19}\mathrm{K} + {}^{0}_{-1}\mathrm{e} \longrightarrow {}^{40}_{18}\mathrm{Ar}$$

The product nucleus has an atomic number that is one less than that of the original nucleus. The mass number remains the same. When another orbital electron fills the vacancy in the inner-shell orbital created by electron capture, an X-ray photon is emitted.

5. **Gamma emission** (abbreviated γ): *emission from an excited nucleus of a gamma photon, corresponding to radiation with a wavelength of about 10^{-12} m.* ◀ In many cases, radioactive decay results in a product nucleus that is in an excited state. As in the case of atoms, the excited state is unstable and goes to a lower-energy state with the emission of electromagnetic radiation. For nuclei, this radiation is in the gamma-ray region of the spectrum.

Recall from Chapter 7 that the energy of a photon can be calculated by the equation $E = hc/\lambda$, where h is Planck's constant, c is the speed of light, and λ is the wavelength of the radiation.

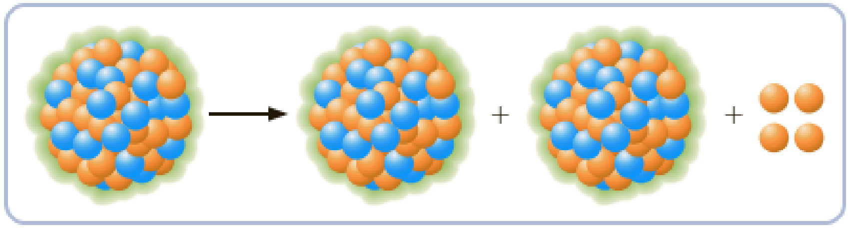
Often gamma emission occurs very quickly after radioactive decay. In some cases, however, an excited state has significant lifetime before it emits a gamma photon. A **metastable nucleus** is *a nucleus in an excited state with a lifetime of at least one nanosecond (10^{-9} s).* In time, the metastable nucleus decays by gamma emission. An example is metastable technetium-99, denoted ${}^{99m}_{43}\text{Tc}$, which is used in medical diagnosis, as discussed in Section 20.5.



$${}^{99m}_{43}\text{Tc} \longrightarrow {}^{99}_{43}\text{Tc} + {}^{0}_{0}\gamma$$

The product nucleus is simply a lower-energy state of the original nucleus, so there is no change of atomic number or mass number.

6. **Spontaneous fission:** *the spontaneous decay of an unstable nucleus in which a heavy nucleus of mass number greater than 89 splits into lighter nuclei and energy is released.* For example, a uranium-236 atom can spontaneously undergo the following nuclear reaction:



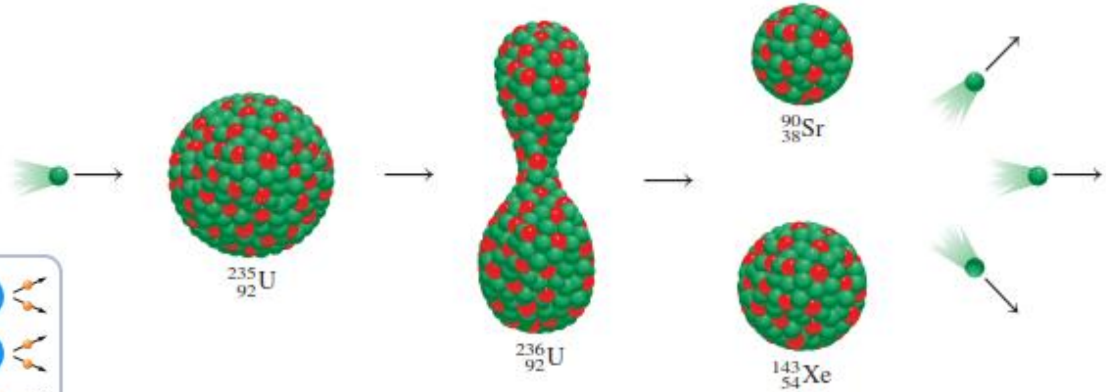
$${}^{236}_{92}\text{U} \longrightarrow {}^{96}_{39}\text{Y} + {}^{136}_{53}\text{I} + 4{}^{1}_{0}\text{n}$$

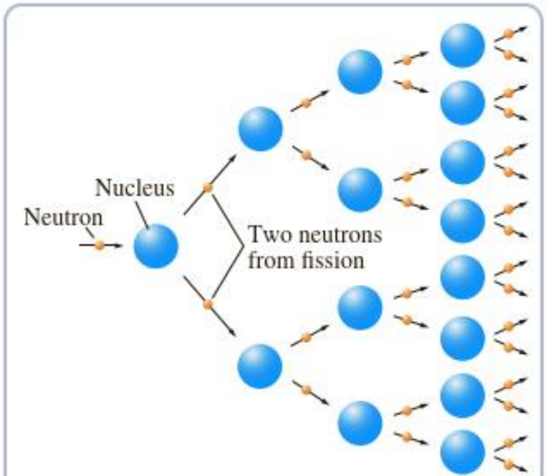
19.5 Nuclear Fission

Nuclear fission is the process in which *a heavy nucleus (mass number > 200) divides to form smaller nuclei of intermediate mass and one or more neutrons.* Because the heavy nucleus is less stable than its products (see Figure 19.2), this process releases a large amount of energy.

The first nuclear fission reaction to be studied was that of uranium-235 bombarded with slow neutrons, whose speed is comparable to that of air molecules at room temperature. Under these conditions, uranium-235 undergoes fission, as shown in

Figure 19.6 *Nuclear fission of ^{235}U. When a ^{235}U nucleus captures a neutron (green sphere), it undergoes fission to yield two smaller nuclei. On the average, 2.4 neutrons are emitted for every ^{235}U nucleus that divides.*





Nuclear Fission

Figure 19.6. Actually, this reaction is very complex: More than 30 different elements have been found among the fission products (Figure 19.7). A representative reaction is

$$^{235}_{92}\text{U} + ^{1}_{0}\text{n} \longrightarrow ^{90}_{38}\text{Sr} + ^{143}_{54}\text{Xe} + 3^{1}_{0}\text{n}$$

Although many heavy nuclei can be made to undergo fission, only the fission of naturally occurring uranium-235 and of the artificial isotope plutonium-239 has any practical importance. Table 19.5 shows the nuclear binding energies of uranium-235 and its fission products. As the table shows, the binding energy per nucleon for uranium-235 is less than the sum of the binding energies for strontium-90 and xenon-143. Therefore, when a uranium-235 nucleus is split into two smaller nuclei, a certain amount of energy is released. Let us estimate the magnitude of this energy. The difference between the binding energies of the reactants and products is $(1.23 \times 10^{-10} + 1.92 \times 10^{-10})$ J – (2.82×10^{-10}) J, or 3.3×10^{-11} J per uranium-235 nucleus. For 1 mole of uranium-235, the energy released would be $(3.3 \times 10^{-11}) \times (6.02 \times 10^{23})$, or 2.0×10^{13} J. This is an extremely exothermic reaction, considering that the heat of combustion of 1 ton of coal is only about 5×10^{7} J.

The significant feature of uranium-235 fission is not just the enormous amount of energy released, but the fact that more neutrons are produced than are originally captured in the process. This property makes possible a ***nuclear chain reaction,*** which is *a self-sustaining sequence of nuclear fission reactions*. The neutrons generated during the initial stages of fission can induce fission in other uranium-235 nuclei, which in turn produce more neutrons, and so on. In less than a second, the reaction can become uncontrollable, liberating a tremendous amount of heat to the surroundings.

For a chain reaction to occur, enough uranium-235 must be present in the sample to capture the neutrons. Otherwise, many of the neutrons will escape from the sample and the chain reaction will not occur. In this situation the mass of the sample is said to be *subcritical*. Figure 19.8 shows what happens when the amount of the fissionable material is equal to or greater than the ***critical mass,*** *the minimum mass of fissionable material required to generate a self-sustaining nuclear chain reaction*. In this case, most of the neutrons will be captured by uranium-235 nuclei, and a chain reaction will occur.

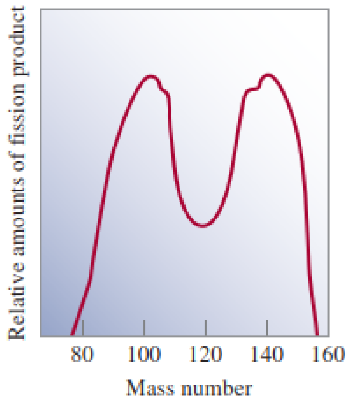


Figure 19.7 *Relative yields of the products resulting from the fission of ^{235}U as a function of mass number.*

Table 19.5

Nuclear Binding Energies of ^{235}U and Its Fission Products

	Nuclear Binding Energy
^{235}U	2.82×10^{-10} J
^{90}Sr	1.23×10^{-10} J
^{143}Xe	1.92×10^{-10} J

The Atomic Bomb

The first application of nuclear fission was in the development of the atomic bomb. How is such a bomb made and detonated? The crucial factor in the bomb's design is the determination of the critical mass for the bomb. A small atomic bomb is equivalent to 20,000 tons of TNT (trinitrotoluene). Because 1 ton of TNT releases about 4×10^9 J of energy, 20,000 tons would produce 8×10^{13} J. Earlier we saw that 1 mole, or 235 g, of uranium-235 liberates 2.0×10^{13} J of energy when it undergoes fission. Thus, the mass of the isotope present in a small bomb must be at least

$$235 \text{ g} \times \frac{8 \times 10^{13} \text{ J}}{2.0 \times 10^{13} \text{ J}} \approx 1 \text{ kg}$$

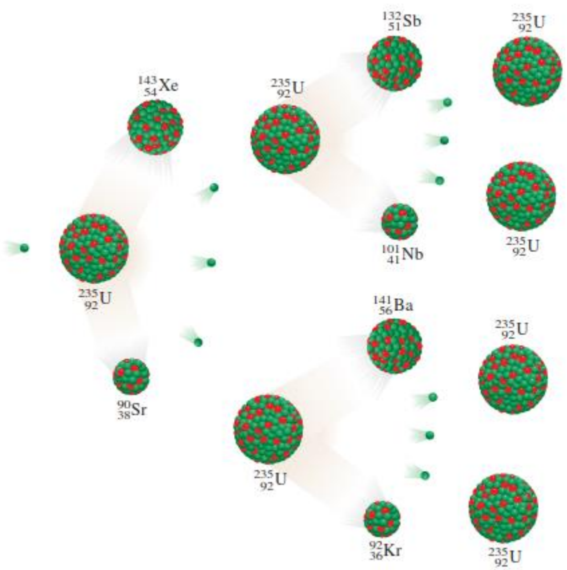


Figure 19.8 *If a critical mass is present, many of the neutrons emitted during the fission process will be captured by other ^{235}U nuclei and a chain reaction will occur.*

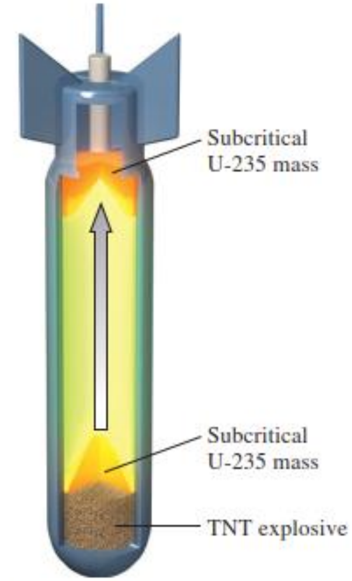


Figure 19.9 *Schematic diagram of an atomic bomb. The TNT explosives are set off first. The explosion forces the sections of fissionable material together to form an amount considerably larger than the critical mass.*

For obvious reasons, an atomic bomb is never assembled with the critical mass already present. Instead, the critical mass is formed by using a conventional explosive, such as TNT, to force the fissionable sections together, as shown in Figure 19.9. Neutrons from a source at the center of the device trigger the nuclear chain reaction. Uranium-235 was the fissionable material in the bomb dropped on Hiroshima, Japan, on August 6, 1945. Plutonium-239 was used in the bomb exploded over Nagasaki three days later. The fission reactions generated were similar in these two cases, as was the extent of the destruction.

19.6 Nuclear Fusion

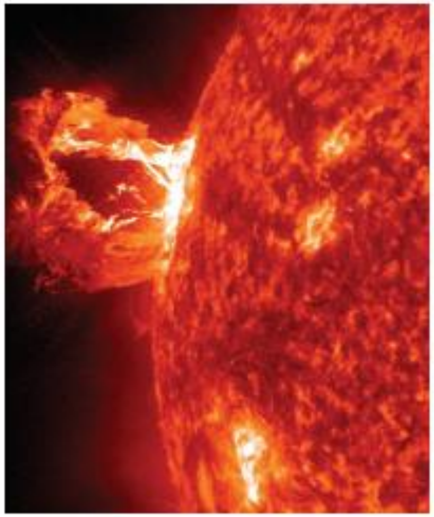
In contrast to the nuclear fission process, ***nuclear fusion,*** *the combining of small nuclei into larger ones,* is largely exempt from the waste disposal problem.

Figure 19.2 showed that for the lightest elements, nuclear stability increases with increasing mass number. This behavior suggests that if two light nuclei combine or fuse together to form a larger, more stable nucleus, an appreciable amount of energy will be released in the process. This is the basis for ongoing research into the harnessing of nuclear fusion for the production of energy.

Nuclear fusion occurs constantly in the sun. The sun is made up mostly of hydrogen and helium. In its interior, where temperatures reach about 15 million degrees Celsius, the following fusion reactions are believed to take place:

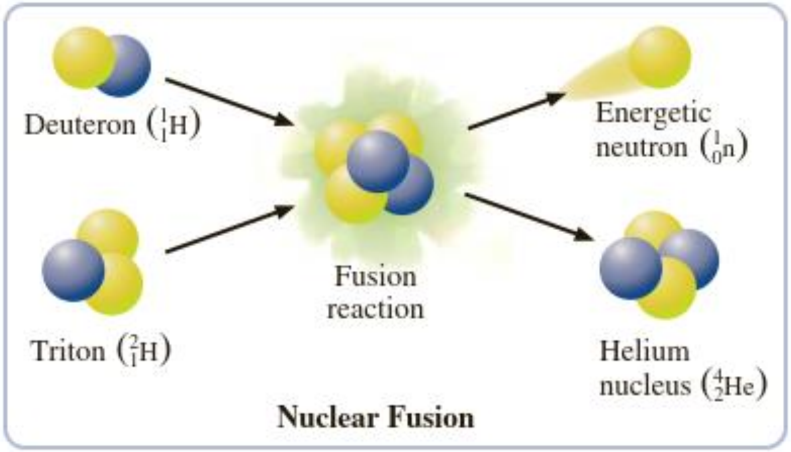
$$^{1}_{1}\mathrm{H} + {}^{2}_{1}\mathrm{H} \longrightarrow {}^{3}_{2}\mathrm{He}$$
$$^{3}_{2}\mathrm{He} + {}^{3}_{2}\mathrm{He} \longrightarrow {}^{4}_{2}\mathrm{He} + 2\,{}^{1}_{1}\mathrm{H}$$
$$^{1}_{1}\mathrm{H} + {}^{1}_{1}\mathrm{H} \longrightarrow {}^{2}_{1}\mathrm{H} + {}^{\;\,0}_{+1}\beta$$

Because *fusion reactions take place only at very high temperatures,* they are often called ***thermonuclear reactions.***



Nuclear fusion keeps the temperature in the interior of the sun at about 15 million °C.

Source: NASA/SDO



The Hydrogen Bomb

The technical problems inherent in the design of a nuclear fusion reactor do not affect the production of a hydrogen bomb, also called a thermonuclear bomb. In this case, the objective is all power and no control. Hydrogen bombs do not contain gaseous hydrogen or gaseous deuterium; they contain solid lithium deuteride (LiD), which can be packed very tightly. The detonation of a hydrogen bomb occurs in two stages—first a fission reaction and then a fusion reaction. The required temperature for fusion is achieved with an atomic bomb. Immediately after the atomic bomb explodes, the following fusion reactions occur, releasing vast amounts of energy (Figure 19.16):

$$^{6}_{3}\text{Li} + ^{2}_{1}\text{H} \longrightarrow 2\,^{4}_{2}\alpha$$
$$^{2}_{1}\text{H} + ^{2}_{1}\text{H} \longrightarrow ^{3}_{1}\text{H} + ^{1}_{1}\text{H}$$

There is no critical mass in a fusion bomb, and the force of the explosion is limited only by the quantity of reactants present. Thermonuclear bombs are described as being "cleaner" than atomic bombs because the only radioactive isotopes they produce are tritium, which is a weak β-particle emitter ($t_{\frac{1}{2}} = 12.5$ yr), and the products of the fission starter. Their damaging effects on the environment can be aggravated, however, by incorporating in the construction some nonfissionable material such as cobalt. Upon bombardment by neutrons, cobalt-59 is converted to cobalt-60, which is a very strong γ-ray emitter with a half-life of 5.2 yr. The presence of radioactive cobalt isotopes in the debris or fallout from a thermonuclear explosion would be fatal to those who survived the initial blast.

Figure 19.16 *Explosion of a thermonuclear bomb.*

Half-Life of a Reaction

As a reaction proceeds, the concentration of a reactant decreases, because it is being consumed. The **half-life**, $t_{1/2}$, of a reaction is *the time it takes for the reactant concentration to decrease to one-half of its initial value.* ▶

The half-life of a reaction was discussed in Section 13.4.

Iodine is attracted to the thyroid gland, which incorporates the element into the growth hormone thyroxine. Radiation from iodine-131 kills cancer cells in the thyroid gland.

We define the **half-life** of a radioactive nucleus as *the time it takes for one-half of the nuclei in a sample to decay.* ◀ The half-life is independent of the amount of sample. In some cases, you can find the half-life of a radioactive sample by directly observing how long it takes for one-half of the sample to decay. For example, you find that 1.000 g of iodine-131, an isotope used in treating thyroid cancer, decays to 0.500 g in 8.07 days. ◀ Thus, its half-life is 8.07 days. In another 8.07 days, this sample would decay to one-half of 0.500 g, or 0.250 g, and so forth. Figure 20.13 shows this decay pattern.

Although you might be able to obtain the half-life of a radioactive nucleus by direct observation in some cases, this is impossible for many nuclei because they decay too quickly or too slowly. Uranium-238, for example, has a half-life of 4.51 billion years, much too long to be directly observed! The usual method of determining the half-life is by measuring decay rates and relating them to half-lives.

You relate the half-life for radioactive decay, $t_{1/2}$, to the decay constant k by the equation

$$t_{1/2} = \frac{0.693}{k}$$

Example 20.8 Calculating the Half-Life from the Decay Constant

Gaining Mastery Toolbox

Critical Concept 20.8
The half-life of a nuclide undergoing nuclear decay is the time it takes for one-half of the nuclei in the sample to decay. The half-life of a sample is independent of the amount of radioactive material and depends only upon the decay constant of the nuclide in the sample.

Solution Essentials:

- Half-life equation for nuclear decay ($t_{1/2} = \frac{0.693}{k}$)
- Half-life
- Radioactive decay constant

The decay constant for the beta decay of $^{99}_{43}Tc$ was obtained in Example 20.7. We found that k equals $1.0 \times 10^{-13}/s$. What is the half-life of this isotope in years?

Problem Strategy Because the decay constant is given in the problem, the preceding equation can be used to calculate the half-life ($t_{1/2}$) from k.

Solution Substitute the value of k into the preceding equation.

$$t_{1/2} = \frac{0.693}{k} = \frac{0.693}{1.0 \times 10^{-13}/s} = 6.9 \times 10^{12}\ s$$

Then you convert this half-life in seconds to years.

$$6.9 \times 10^{12}\ s \times \frac{1\ min}{60\ s} \times \frac{1\ h}{60\ min} \times \frac{1\ d}{24\ h} \times \frac{1\ y}{365\ d} = \mathbf{2.2 \times 10^5\ y}$$

Radioactive Dating

Fixing the dates of relics and stone implements or pieces of charcoal from ancient campsites is an application based on radioactive decay rates. Because the rate of radioactive decay of a nuclide is constant, this rate can serve as a clock for dating very old rocks and human implements. Dating wood and similar carbon-containing objects that are several thousand to fifty thousand years old can be done with radioactive carbon, carbon-14, which has a half-life of 5730 y.

Carbon-14 is present in the atmosphere as a result of cosmic-ray bombardment of earth. Cosmic rays are radiations from space that consist of protons and alpha particles, as well as heavier ions. These radiations produce other kinds of particles, including neutrons, as they bombard the upper atmosphere. The collision of a neutron with a nitrogen-14 nucleus (the most abundant nitrogen nuclide) can produce a carbon-14 nucleus.

$$^{14}_{7}N + ^{1}_{0}n \longrightarrow ^{14}_{6}C + ^{1}_{1}H$$

Carbon dioxide containing carbon-14 mixes with the lower atmosphere. Because of the constant production of $^{14}_{6}C$ and its radioactive decay, a small, constant fractional abundance of carbon-14 is maintained in the atmosphere.

Living plants, which continuously use atmospheric carbon dioxide, also maintain a constant abundance of carbon-14. Similarly, living animals, by feeding on plants, have a constant fractional abundance of carbon-14. But once an organism dies, it is no longer in chemical equilibrium with atmospheric CO_2. The ratio of carbon-14 to carbon-12 begins to decrease by radioactive decay of carbon-14. In this way, this ratio of carbon isotopes becomes a clock measuring the time since the death of the organism.

If you assume that the ratio of carbon isotopes in the lower atmosphere has remained at the present level for the last 50,000 years (presently 1 out of 10^{12} carbon atoms is carbon-14), you can deduce the age of any dead organic object by measuring the level of beta emissions that arise from the radioactive decay of carbon-14. ▶ This is the decay reaction:

$$^{14}_{6}C \longrightarrow ^{14}_{7}N + ^{0}_{-1}e$$

In this way, bits of campfire charcoal, parchment, jawbones, and the like have been dated.

Analyses of tree rings have shown that this assumption is not quite valid. Before 1000 B.C., the levels of carbon-14 were somewhat higher than they are today. Moreover, recent human activities (burning of fossil fuels and atmospheric nuclear testing) have changed the fraction of carbon-14 in atmospheric CO_2.

Isotopes in Medicine

Tracers are used also for diagnosis in medicine. Sodium-24 (a β emitter with a half-life of 14.8 h) injected into the bloodstream as a salt solution can be monitored to trace the flow of blood and detect possible constrictions or obstructions in the circulatory system. Iodine-131 (a β emitter with a half-life of 8 days) has been used to test the activity of the thyroid gland. A malfunctioning thyroid can be detected by giving the patient a drink of a solution containing a known amount of $Na^{131}I$ and measuring the radioactivity just above the thyroid to see if the iodine is absorbed at the normal rate. Of course, the

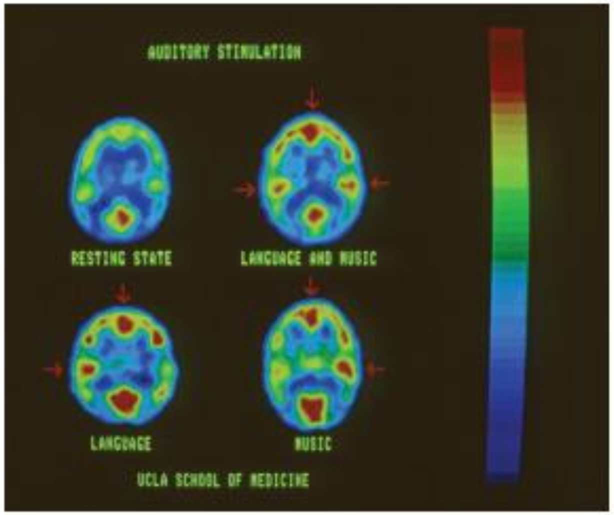


Figure 19.17 *This patient was injected with a small dose of a radioactive tracer that binds to glucose molecules in the blood. The glucose concentrates in the more active regions of the brain that can then be identified by gamma emission (shown as red in the figure), allowing one to determine which parts of the brain are associated with different activities.*

amounts of radioisotope used in the human body must always be kept small; otherwise, the patient might suffer permanent damage from the high-energy radiation. A radioactive isotope of fluorine, fluorine-18, emits positrons, which are annihilated by electrons, forming γ-rays used to image the brain (Figure 19.17). Table 19.6 shows some of the radioactive isotopes used in medicine.

Technetium, the first artificially prepared element, is one of the most useful elements in nuclear medicine. Although technetium is a transition metal, all its isotopes are radioactive. In the laboratory it is prepared by the nuclear reactions

$$^{98}_{42}\mathrm{Mo} + ^{1}_{0}\mathrm{n} \longrightarrow ^{99}_{42}\mathrm{Mo}$$
$$^{99}_{42}\mathrm{Mo} \longrightarrow ^{99\mathrm{m}}_{43}\mathrm{Tc} + ^{0}_{-1}\beta$$

where the superscript m denotes that the technetium-99 isotope is produced in its excited nuclear state. This isotope has a half-life of about 6 h, decaying by γ radiation to technetium-99 in its nuclear ground state. Thus, it is a valuable diagnostic tool. The patient either drinks or is injected with a solution containing $^{99\mathrm{m}}\mathrm{Tc}$. By detecting the γ rays emitted by $^{99\mathrm{m}}\mathrm{Tc}$, doctors can obtain images of organs such as the heart, liver, and lungs.

A major advantage of using radioactive isotopes as tracers is that they are easy to detect. Their presence even in very small amounts can be detected by photographic techniques or by devices known as counters. Figure 19.18 is a diagram of a Geiger counter, an instrument widely used in scientific work and medical laboratories to detect radiation.

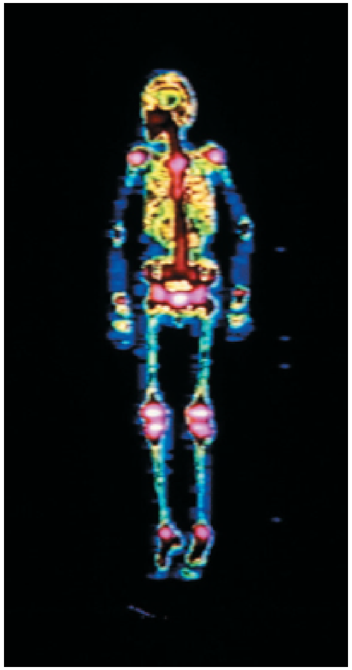


Image of a person's skeleton obtained using $^{99\mathrm{m}}_{43}\mathrm{Tc}$.

Table 19.6 Some Radioactive Isotopes Used in Medicine

Isotope	Half-Life	Uses
$^{18}\mathrm{F}$	1.8 h	Brain imaging, bone scan
$^{24}\mathrm{Na}$	15 h	Monitoring blood circulation
$^{32}\mathrm{P}$	14.3 d	Location of ocular, brain, and skin tumors
$^{43}\mathrm{K}$	22.4 h	Myocardial scan
$^{47}\mathrm{Ca}$	4.5 d	Study of calcium metabolism
$^{51}\mathrm{Cr}$	27.8 d	Determination of red blood cell volume, spleen imaging, placenta localization
$^{60}\mathrm{Co}$	5.3 yr	Sterilization of medical equipment, cancer treatment
$^{99\mathrm{m}}\mathrm{Tc}$	6 h	Imaging of various organs, bones, placenta location
$^{125}\mathrm{I}$	60 d	Study of pancreatic function, thyroid imaging, liver function
$^{131}\mathrm{I}$	8 d	Brain imaging, liver function, thyroid activity

Hazards of Nuclear Energy

Many people, including environmentalists, regard nuclear fission as a highly undesirable method of energy production. Many fission products such as strontium-90 are dangerous radioactive isotopes with long half-lives. Plutonium-239, used as a nuclear fuel and produced in breeder reactors, is one of the most toxic substances known. It is an alpha emitter with a half-life of 24,400 yr.

Accidents, too, present many dangers. An accident at the Three Mile Island reactor in Pennsylvania in 1979 first brought the potential hazards of nuclear plants to public attention. In this instance, very little radiation escaped the reactor, but the plant remained closed for more than a decade while repairs were made and safety issues addressed. Only a few years later, on April 26, 1986, a reactor at the Chernobyl nuclear plant in Ukraine surged out of control. The fire and explosion that followed released much radioactive material into the environment. People working near the plant died within weeks as a result of the exposure to the intense radiation. Twenty-five years later, it is still not known how many people died from radiation-induced cancer cases. Estimates vary between a few thousand and hundreds of thousands. The latest large-scale nuclear plant accident occurred in Fukushima, Japan, on March 11, 2011. A powerful earthquake, followed by a destructive tsunami, severely damaged the nuclear reactors at the plant. The long-term harmful effects of the radiation leakage to the environment are not yet fully assessed, but it is believed to be comparable to that at Chernobyl.

In addition to the risk of accidents, the problem of radioactive waste disposal has not been satisfactorily resolved even for safely operated nuclear plants. Many suggestions have been made as to where to store or dispose of nuclear waste, including burial underground, burial beneath the ocean floor, and storage in deep geologic formations. But none of these sites has proved absolutely safe in the long run. Leakage of radioactive wastes into underground water, for example, can endanger nearby communities. The ideal disposal site would seem to be the sun, where a bit more radiation would make little difference, but this kind of operation requires 100 percent reliability in space technology.

Because of the hazards, the future of nuclear reactors is clouded. What was once hailed as the ultimate solution to our energy needs in the twenty-first century is now being debated and questioned by both the scientific community and laypeople. It seems likely that the controversy will continue for some time.